

Socioeconomic impacts of Terrestrial Protected Areas

Evidence from Large-Scale National Surveys in Madagascar

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Introduction

To what extent do protected areas impact household living standards?

► **Economic Opportunities**

- Development of local tourism (Naidoo et al. 2019),
- Creation of conservation-related jobs (Balietti et al. 2024)
- Compensation measures (Poudyal et al. 2018)
- Improved ecosystem services (Xu et al. 2023)

► **Restrictions and Costs** Restricted access to natural resources - gathering, hunting, fishing, medicinal plants (Kandel et al. 2022)

► **Heterogeneous results** Effects vary across evaluation methods, institutional contexts, and geographic locations (Kandel et al. 2022).

Why Madagascar?

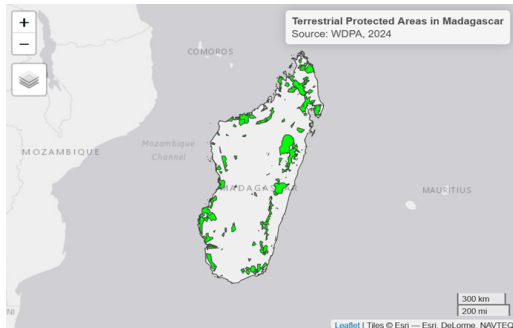
Why Madagascar?

The area of protected areas and the size of the adjacent population exposed to them nearly tripled between 2008 and 2021

- ▶ 2008 : 3,6% of the territory — 9% population lived < 10 km of protected area
- ▶ 2021 : 10,8% du territoire — 28% population lived > 10 km of protected area

A Vulnerable socio-economic context

- ▶ The world's poorest country based on SDG Indicator 1.1 (Conceição 2024)
- ▶ High dependence on natural resources (Llopis et al. 2023)
- ▶ Weak state capacity (Hanson and



Source : Authors

Theory of Change

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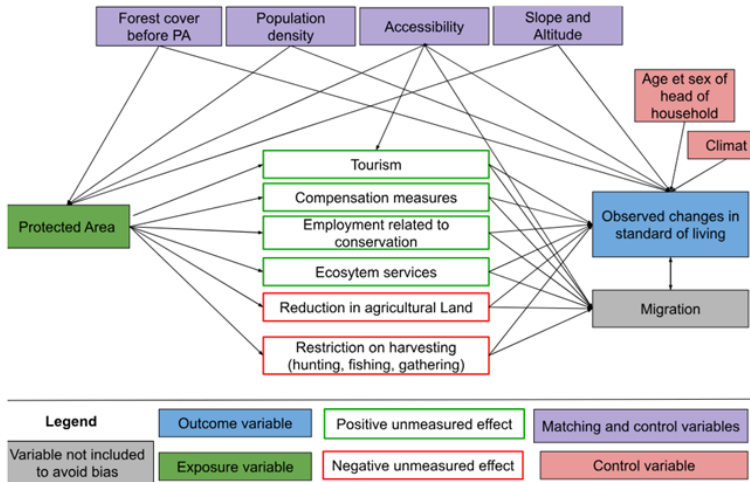


Figure 1: Logic diagram of the theory of change

Empirical methodology

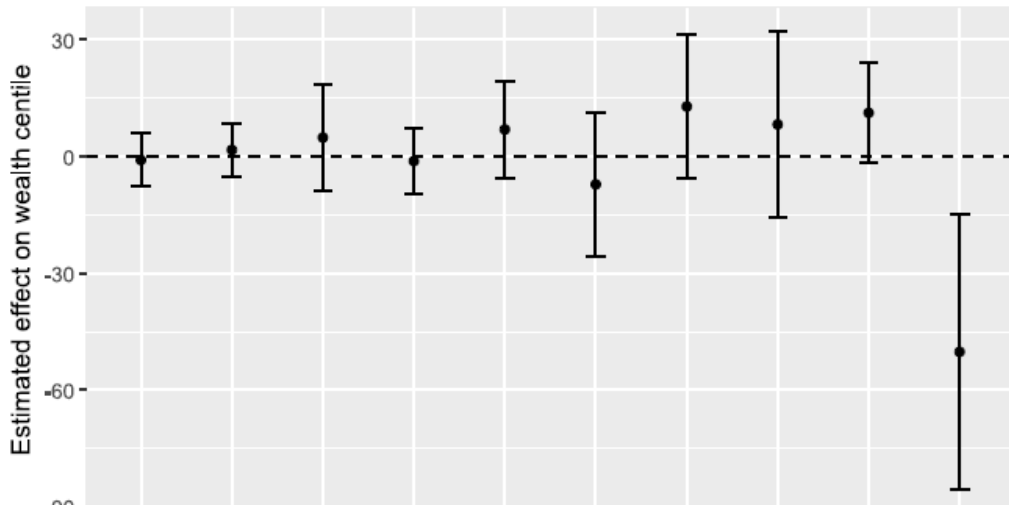
The data at the rural level

- ▶ Combine geolocated household data from six nationally representative survey waves Demographic and Health surveys (1997, 2008, 2021), Malaria Indicator Surveys (2011, 2013, 2016) and Multiple Indicator Cluster Surveys (2018) - with the World Database on Protected Areas and geospatial environmental covariates (forest cover, slope, elevation, population density, accessibility, and standardized precipitation evapotranspiration index).
- ▶ **Dependent variable:** Protected Areas Implementation, from WDPA, SAPM
- ▶
- ▶ **Treatment group:** households living in rural areas < 10 km of a Protected areas created between 2008 and 2021
- ▶ **Control group:** households living in rural areas < 10 km of a Protected areas created between 2008 and 2021

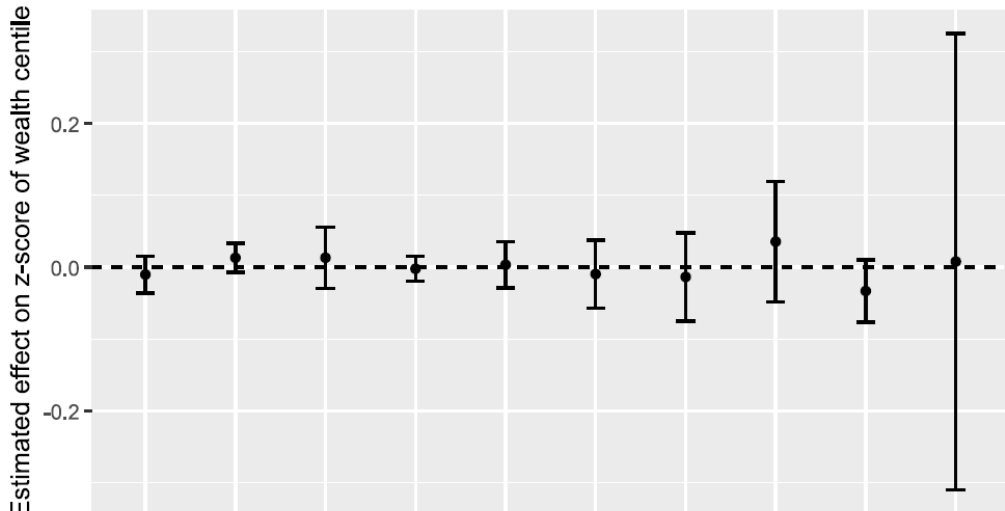
Baseline model: protected areas, living conditions, and rural

Results

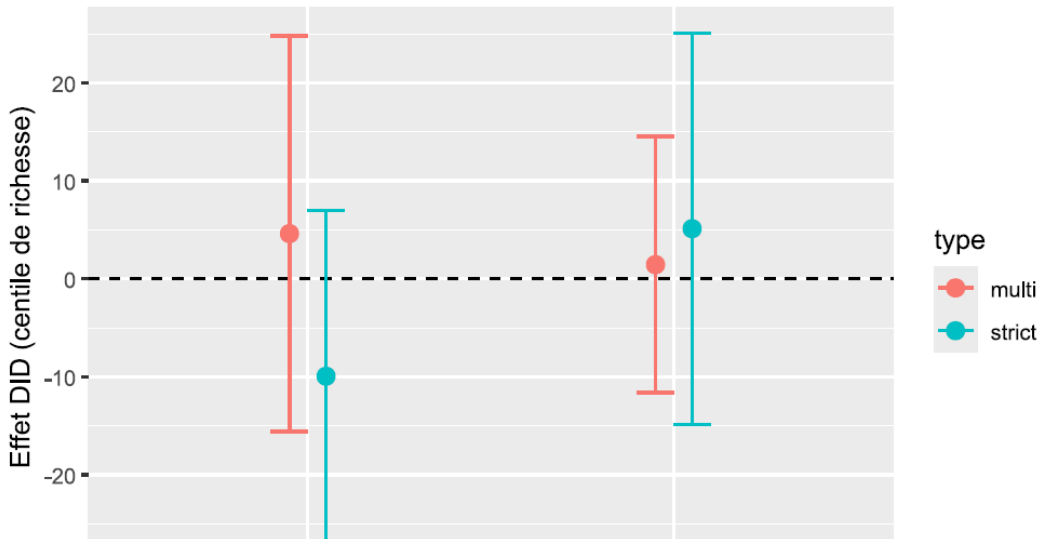
Protected Areas impact on the household wealth index



Protected Areas impact on the household wealth index inequalities



Heterogeneity: Which Protected Areas Harm Most?



Policy Implications

References

- Balietti, Anca, Sreeja Jaiswal, and Daniel Schäffer. 2024. "Labor Market Impacts of Eco-Development Initiatives in Protected Areas." *Journal of Environmental Economics and Management* 128 (November): 103070. <https://doi.org/10.1016/j.jeem.2024.103070>.
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- Kandel, Pratikshya, Ram Pandit, Benedict White, and Maksym Polyakov. 2022. "Do Protected Areas Increase Household Income? Evidence from a Meta-Analysis"